How I Approached the Problem | Lead Quality Analysis | Debt Reduction Case Study

THE DATA

The dataset contained 3,021 leads sold to a single advertiser in the debt reduction space. Each lead had a disposition status, plus metadata on the ad creative shown, traffic source, consumer debt level, state, and verification scores.

DATA CLEANUP

1. Partner names were inconsistent ("google" vs "Google" appeared as separate entries). Merged them.
2. Widget sizes 300x250 and 302x252 were identical creatives per the documentation but coded differently. Unified them.
4. Address and phone scores only existed for about 40-46% of leads since they were rolled out mid-period. Usable where available, but not reliable for full-dataset analysis.
5. IP addresses were removed from the dataset, so any geo-validation or home/work resolution was off the table.

DIFFICULTIES AND TRADE-OFFS

1. 70.8% of leads sat in "Unknown" status with no clear good or bad outcome. I kept them in the denominator (intent-to-treat) so quality rates reflect reality rather than a cherry-picked subset, but this means the true conversion picture could shift if those leads resolve unevenly.
2. Phone and address scores were the most promising predictive signals, but with data on fewer than half the leads, any findings there come with a sample size caveat.
3. High-cardinality fields like keywords and referral URLs had too few leads per value to draw meaningful conclusions, so I deprioritized them in favor of dimensions with enough volume to test properly.
4. Ad Zone and Channel turned out to be the same variable (TopLeft/TopRight maps directly to Online/Call Center). I reported it once under Channel to avoid double-counting the same insight.

HOW I DEFINED QUALITY

I followed the advertiser's own framework. Close rate (Closed / Total) is the primary metric since it is what the advertiser ultimately cares about. I also tracked a good lead rate (Closed + EP stages) and a bad lead rate (Unable to Contact + Invalid Profile + Doesn't Qualify) as supporting signals.

APPROACH

Question 1: Is quality trending?

I plotted close rate by week and month, then ran a regression on weekly data to test significance. Quality is declining (10.8% in April to 4.4% in September, $p = 0.008$). I then decomposed why: the mix of incoming leads shifted. Low-debt leads under $10K grew from nearly zero to 21% of volume, and that tier converts the worst. The decline is partly structural, not just performance degradation.

Question 2: What drives quality?

I Ran chi-square tests across every dimension. Ranked by quality spread and significance. Debt level is the clear #1 driver ($p = 0.006$). BlueMeter creative and Google campaign selection are secondary levers.

Question 3: Can we hit the 9.6% target?

I modeled filtering scenarios: what happens to close rate and volume if we cut the lowest-quality segments. Built conservative, moderate, and aggressive options, each showing the resulting quality, volume trade-off, and impact on the advertiser's cost per acquisition. The goal was to present clear trade-offs, not a single recommendation.

VISUALIZATION LOGIC

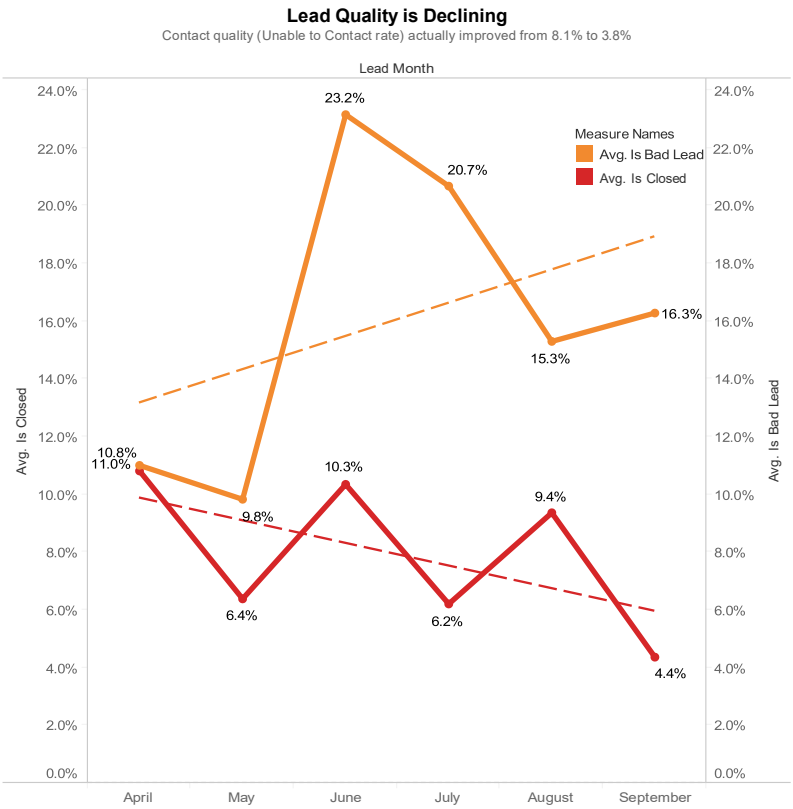
Every chart answers one question. The trend chart shows trajectory with a regression line so the decline is immediate. The driver charts rank segments by quality spread so you see what matters most at a glance. The scenario table puts volume, quality, and revenue side by side so the trade-offs are transparent. Nothing decorative. Everything earns its place by making a decision easier.

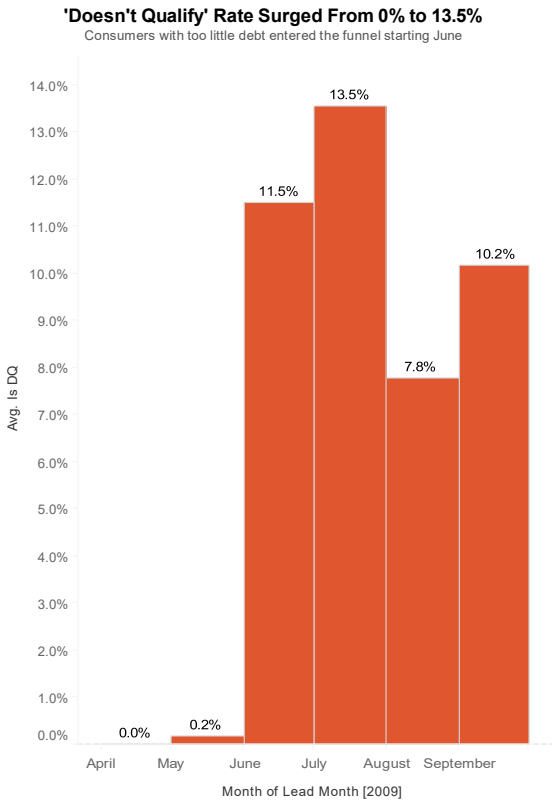
References: Tableau Dashboard: Link
Data Analysis: Link

The full analysis and visualizations follow on the next pages.

Lead Quality is Declining - Driven by Audience Mix Changes, Not Lead Source Quality

Q1: Are we seeing any lead quality trends over time?
Are they statistically significant?



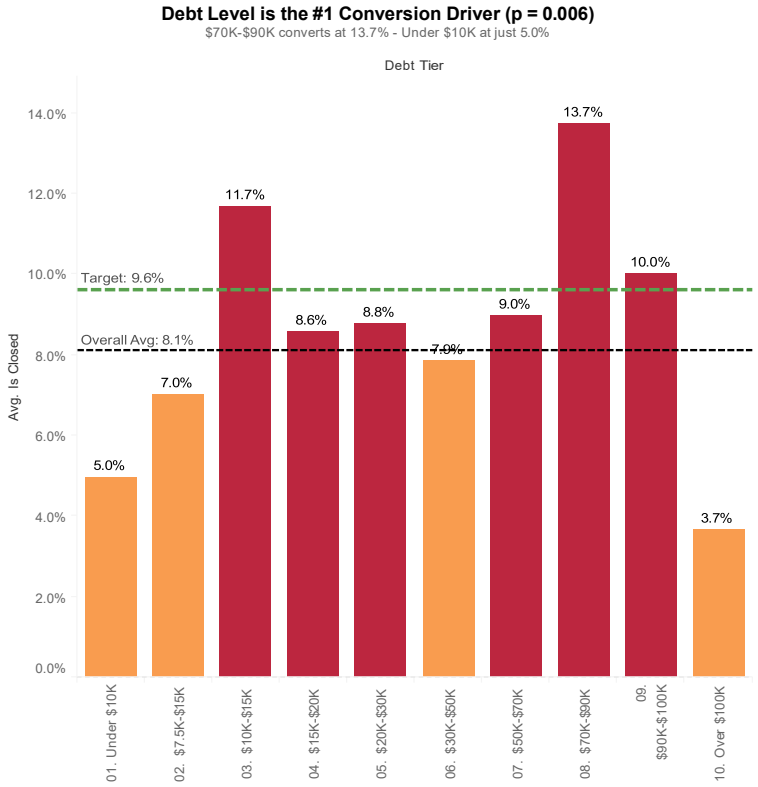


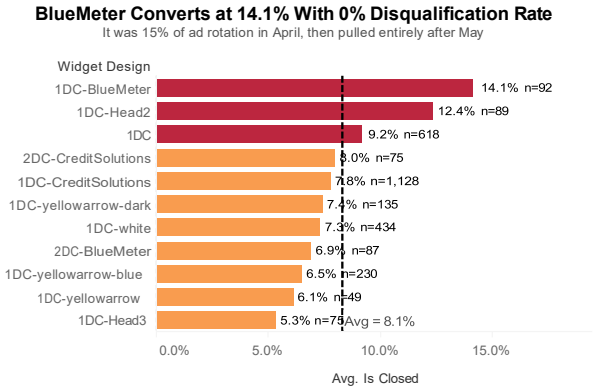
FINDINGS

1. Lead quality is declining over time, with conversion dropping from 10.8% (April) to 4.4% (September).
2. This decline is statistically significant ($p = 0.008$), confirming it is not due to random variation.
3. The primary driver is a shift in audience composition toward lower-quality users.
4. Users with less than $10K debt increased from 0% to 22% and convert at ~5%.
5. High-performing creatives were removed and replaced with lower-performing ones.
6. Traffic shifted toward Google, replacing higher-converting partners.
7. Contactability improved (8.1% to 3.8%), so lead validity is not the issue
8. Overall, the decline is driven by targeting and campaign mix changes.

Debt Level, Ad Creative, and Google Campaign Selection Drive Lead Quality

Q2: What can we learn about the drivers of "lead quality" from this dataset?
What segments have differing lead quality rates?



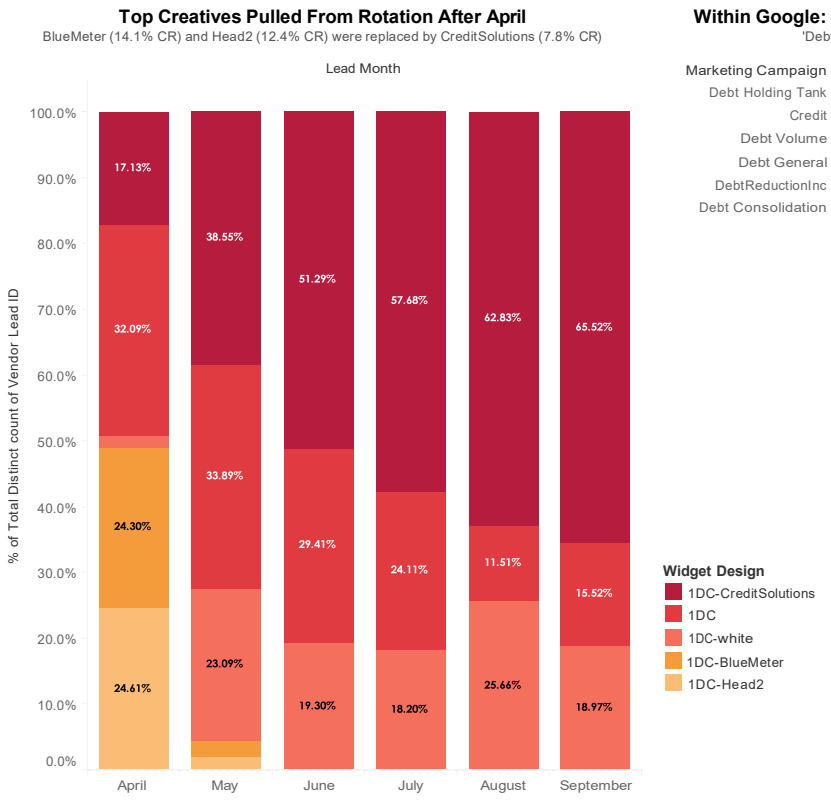


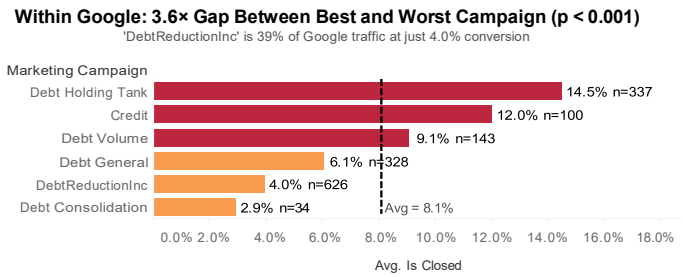
FINDINGS

1. Lead quality varies significantly based on user characteristics, ad creative, and campaign selection.
2. Debt level is the strongest and only statistically significant driver (p = 0.006).
3. Users with $70K–$90K debt convert at 13.7%, while users under $10K convert at ~5%.
4. The optimal segment is users with $10K–$100K debt.
5. Ad creative has a large impact, with a ~9 percentage point difference across creatives.
6. BlueMeter performs best (14.1%) and attracts higher-quality users.
7. Campaign performance varies widely, with a 3.6x gap between best and worst.
8. “Debt Holding Tank” converts at 14.5%, while “DebtReductionInc” converts at 4.0%.
9. Factors such as branding, form design, and channel have minimal impact.
10. Overall, lead quality is driven by user segment, creative, and campaign choice.

Two Paths to 9.6%+ Conversion - Both Using Standard Campaign Optimization

Q3: If the advertiser increases CPL by 20% for a 20% quality increase, do we see opportunities to do that?





RECOMMENDED PATHS

A 20% improvement in conversion (8.0% to 9.6%) is achievable through focused optimization.

This can be achieved by focusing on higher-quality segments and optimizing campaigns.

Path 1: Target users with $10K - $100K debt and remove low-performing creatives.
This increases conversion to 9.6% with ~47% reduction in volume.

Path 2: Reallocate budget from low-performing to high-performing Google campaign.
Pause “DebtReductionInc” (4.0%) and invest in “Debt Holding Tank” (14.5%).
This increases conversion to 10.3% with ~50% reduction in volume.

Additional recommendation: Introduce tiered CPL pricing to align incentives with incremental quality improvements.

Overall, improving lead quality requires prioritizing efficiency and targeting precision over scale.